MUHANNAD YAZBAK

Full-Stack Developer & QA Automation Engineer

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PROFESSIONAL SUMMARY

Dynamic Software Engineer holding dual B.Sc. degrees, specializing in full-stack web architecture, AI microservices, and end-to-end QA automation. Proven track record building responsive Next.js applications, multi-tenant SaaS architectures, containerized FastAPI services, and native desktop frameworks. Expert in implementing robust Playwright test suites (POM) and optimizing CI/CD pipelines.

TECHNICAL SKILLS

- **Languages & Core:** JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3, REST APIs
- **Frontend & Desktop:** React.js, Next.js, Electron, Vite, Tailwind CSS, Dynamic SEO
- **Backend & AI:** Node.js, Express.js, FastAPI, RAG Microservices, LangChain, OpenRouter API, Resend
- **Databases & ORM:** PostgreSQL, SQLAlchemy, MySQL, MongoDB, SQLite, Mongoose, DBeaver
- **Automation & Testing:** Playwright (POM), Allure Reporting, Selenium, Vitest, Mockoon, E2E Testing
- **DevOps & Cloud:** Docker, Kubernetes (kind), GitHub Actions CI/CD, Vercel, Render, Aiven

CORE PROJECTS

Multi-Tenant Vertical CRM Platform | *Next.js, TypeScript, Python, FastAPI, PostgreSQL, Playwright, Resend*

- **Architecture & Multi-Tenancy:** Engineered a multi-tenant SaaS platform featuring vertical-specific workflows (Legal/Insurance), role-based access control (Admin vs. Tenant), and secure token-based password recovery via Resend.
- **Data Layer & API:** Designed a dynamic PostgreSQL database schema using SQLAlchemy ORM to manage dynamic client tabs (Notes, Documents, Invoicing) with soft-delete capabilities.
- **CI/CD & QA Automation:** Integrated a zero-flake Playwright (POM) test suite executed within GitHub Actions CI/CD pipelines to validate complete multi-role user journeys.

TechMart – Full-Stack E-Commerce Ecosystem | *Next.js, TypeScript, Python, FastAPI, Docker, MySQL*

- **Multi-Cloud Architecture:** Engineered a decoupled web platform separating an edge UI layer (Vercel) from containerized processing loops (Render) and a managed MySQL database (Aiven).
- **AI RAG Microservice:** Built an asynchronous Python RAG service using FastAPI and LangChain to execute real-time semantic search queries across multilingual datasets.
- **Automated Quality Assurance:** Authored a comprehensive Playwright end-to-end framework using the Page Object Model (POM) to validate checkout flows.

Autonomous Playwright Self-Healing Test Agent | *Next.js, FastAPI, Python, Playwright, LLM APIs, GitHub*

- **Self-Healing Engine:** Developed an AI-driven test system that intercepts broken DOM locators at runtime and passes context to an LLM for real-time selector repair.
- **Code Generation:** Built background scripts observing automated exploratory sessions to auto-generate native, structured spec.ts test files.
- **CI/CD Optimization:** Optimized GitHub Actions workflows using custom dependency caching strategies to accelerate test suite execution times.

Localized Insurance Agency ERP (Desktop) | *Electron, Vite, React.js, TypeScript, SQLite, Tailwind CSS*

- **Local-First Architecture:** Built a secure desktop CRM/ERP using Electron and SQLite to manage offline client records, policy lifecycles, and insurance claims.
- **Secure Document Ingestion:** Implemented a custom drag-and-drop ingestion module archiving media to local OS paths via secure Node.js fs handlers.

EDUCATION & TRAINING

- **2023 | QA Automation Engineering Program** – Appleseeds (Full E2E suite for Rami-Levy) |
- **2019–2023 | B.Sc. in Information Systems (BI Specialization)** – University of Haifa
- **2021–2022 | Software Engineering Training** – Tsofen (Java, Spring Boot) & WebAhead (React, JS)
- **2010–2013 | B.Sc. in Medical Laboratory Sciences** – Ben-Gurion University of the Negev (GPA: 88)